Q-Non-Intersecting Lattice Paths

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Abstract

We develop the Q-enriched lattice path model for cylindric plane partitions. In the standard model [5], CPPs biject with families of non-intersecting lattice paths on the cylinder. We define a *Q-lattice path family* as a pair $(S, \mathfrak{p})$, where $\mathfrak{p}$ is a non-intersecting family and S records crossing positions. When $S = \emptyset$ the family is non-intersecting and corresponds to a CPP; when $S \neq \emptyset$ the paths cross and the family carries sign $(-1)^{|S|}$. The positivity of $Q_{n,c}(q)$ becomes the statement that all intersecting families can be cancelled by a sign-reversing involution, leaving only non-intersecting (CPP) contributions.

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1 The standard NILP model

We recall the bijection between cylindric plane partitions and non-intersecting lattice paths (NILPs) on the cylinder [5].

1.1 Paths on the cylinder

A *lattice path* on a cylinder of period T is a sequence $p = (y_0, y_1, \ldots, y_T)$ of integers with $y_{k+1} - y_k \in \{0, 1\}$ for each k. Each step is either:

- **Up**: $y_{k+1} = y_k + 1$ (encoded as 1),
- **Flat**: $y_{k+1} = y_k$ (encoded as 0).

The path is encoded by the binary string $p = (p_0, \ldots, p_{T-1})$ and a starting height y_0.

1.2 Non-intersecting families

Definition 1 (NILP family)**.** A *family of non-intersecting lattice paths* on the cylinder of period T is a collection of paths $p^1, p^2, \ldots, p^m$ satisfying:

1. $y_k^{i+1} > y_k^i$ for all k and i (non-intersecting),
2. $y_0^{i+1} - y_0^i = y_T^{i+1} - y_T^i$ for all i (cylindric closure).

1.3 The CPP–NILP bijection

Theorem 2 ([5], Theorem 4.2)**.** *Let π be a binary string of length T. There is a bijection between cylindric plane partitions of profile π and minimal families of non-intersecting lattice paths on the cylinder of period T with profile π.*

The bijection is given by the column-length encoding*: the i-th path encodes the i-th column length across successive partitions:*

$$\boxed{p_i(\mathfrak{c})_k = (\mu^{k+1})'_i - (\mu^k)'_i + \pi_k \ \in \ \{0,1\},} \tag{1}$$

where $\mathfrak{c} = (\mu^0, \mu^1, \ldots, \mu^T)$ is the interlacing sequence and $(\mu^k)'_i$ is the length of the i-th column of μ^k.

1.4 Cubes

Definition 3 (Cube)**.** A *cube* in the NILP model is a pair of vertices $v_1 = (x, y_1)$ and $v_2 = (x, y_2)$ at the same x-coordinate with $y_1 < y_2$, where v_1 is *occupied* (lies on some path p^i) and v_2 is *unoccupied* (lies between paths p^j and p^{j+1} for some $j \geq i$).

In the standard model, each cube carries weight 1, and the total weight of a NILP family is the number of cubes. This equals the total weight $|\mathfrak{c}|$ of the corresponding CPP.

1.5 Three models

We now have three equivalent models for the same objects:

Model	**Object**	**Local structure**
Interlacing sequence	$(\mu^0, \mu^1, \ldots, \mu^T)$	horizontal strips μ^k/μ^{k-1}
Growth diagram	ALCD with face labels	local rule at each face
Lattice paths	non-intersecting paths	cubes (occupied–unoccupied pairs)

The growth diagram and lattice path models are *dual*: the growth diagram operates on *rows* of the partition (via Pieri operators P_m, T_n), while the lattice paths track *columns*. The cube in the NILP model corresponds to the face label in the growth diagram.

2 Q-weighted lattice paths

We now enrich the NILP model with Q-weights.

2.1 Strip sizes in the NILP picture

At step k of the interlacing sequence $\mathfrak{c} = (\mu^0, \mu^1, \ldots, \mu^T)$, the horizontal strip μ^k/μ^{k-1} (or μ^{k-1}/μ^k, depending on π_k) has size $a_k = ||\mu^k| - |\mu^{k-1}||$.

In the NILP picture, the strip size a_k equals the number of paths that take an "up" step at position k minus the number expected from the profile:

$$a_k = \#\{i : p_i(\mathfrak{c})_k = 1\} - \pi_k \cdot (\text{number of paths}). \tag{2}$$

More precisely, a_k is the number of *free* up-steps at position k (those not forced by the profile).

2.2 Q-weight of a cube

Definition 4 (Q-cube weight)**.** Let $c = (v_1, v_2)$ be a cube at x-coordinate k, with v_1 on path p^i and v_2 between paths p^j and p^{j+1}. The *Q-weight* of c depends on the strip sizes at position k:

$$\mathrm{wt}_Q(c) = Q_{a_k(c)}(q), \tag{3}$$

where $a_k(c)$ is the strip size at position k as seen from the cube c (i.e., the local strip size involving the paths between p^i and p^{j+1}).

In the standard model, $Q_n = 1$ for all n (i.e., $d = 1$), so all cubes carry weight 1. For $d \geq 2$, the Q-weight depends on the geometric context of the cube within the path family.

2.3 Q-weighted NILP families

Definition 5 (Q-NILP)**.** A *Q-weighted non-intersecting lattice path family* (or Q-NILP) is a standard NILP family where each cube c carries the Q-weight (3). The total Q-weight of the family is the product over all cubes:

$$\mathrm{wt}_Q(\mathfrak{c}) = \prod_{\text{cubes } c \text{ in } \mathfrak{c}} Q_{a(c)}(q). \tag{4}$$

3 The Q-kernel as a two-path interaction

The lattice path picture gives a geometric interpretation of the Q-kernel $\Phi_{a,b}(t)$.

3.1 Two adjacent paths

Consider two adjacent paths p^i and p^{i+1} in a NILP family. At each x-coordinate k, each path takes an up or flat step. There are four possibilities:

p^i_k	p^{i+1}_k	**Interaction**
0 (flat)	0 (flat)	paths maintain gap; no cube created
1 (up)	1 (up)	paths maintain gap; no cube created
0 (flat)	1 (up)	gap increases by 1; no cube created
1 (up)	0 (flat)	gap decreases by 1; **cube created**

A cube between p^i and p^{i+1} is created precisely when p^i goes up and p^{i+1} stays flat. In the growth diagram language, this corresponds to a face with label $k \geq 1$: the cascade parameter counts the number of "crossings" where the lower path rises into the territory of the upper path.

3.2 The cascade as gap reduction

At cascade level k, the two-path interaction has the lower path taking k more up-steps than the upper path over the same interval. Each such up-step creates a cube. The Q-weight of the j-th cube is $Q_{a+j}(q)$ (for the lower path) times $Q_{b+j}(q)$ (for the upper path), where a and b are the initial strip sizes.

The Q-kernel is the generating function for these interactions:

$$\boxed{\Phi_{a,b}(t) = \sum_{k \geq 0} Q_{a+k}\, Q_{b+k}\, t^k = \sum_{\substack{\text{two-path} \\ \text{interactions}}} \mathrm{wt}_Q \cdot t^{\#\text{cubes}}.} \tag{5}$$

3.3 The shifted factorisation, geometrically

The shifted factorisation $\Phi_{a,b}(t) = Q_a\, Q_b + t\, \Phi_{a+1,b+1}(t)$ has a geometric meaning in the NILP picture:

- $Q_a \cdot Q_b$: no cubes between the two paths ($k = 0$). The paths maintain their gap throughout. The Q-weight is the product of the individual path weights.
- $t \cdot \Phi_{a+1,b+1}(t)$: at least one cube is created ($k \geq 1$). The first cube has Q-weight $Q_{a+1} \cdot Q_{b+1}$, and the remaining cubes contribute $\Phi_{a+1,b+1}(t)$. The factor t records the first cube.

Peeling off one cube at a time is exactly the layer-by-layer structure of the peel/glue involution ([3], Section 7).

4 Q-lattice path families

We now define the central object: a Q-lattice path family. This is a pair $(S, \mathfrak{p})$ — not just a NILP family $\mathfrak{p}$ — and the subset S records where paths *cross*.

4.1 Pairs as path families with crossings

Definition 6 (Q-lattice path family)**.** A *Q-lattice path family* of level n and profile c is a pair $(S, \mathfrak{p})$ where:

- $\mathfrak{p}$ is a NILP family of $n - |S|$ non-intersecting paths on the cylinder,
- $S \subseteq \{1, \ldots, n\}$ is the set of *crossing positions*,
- sign: $(-1)^{|S|}$,
- weight: $q^{\sigma(S)+|\mathfrak{p}|}$ where $\sigma(S) = \sum_{i \in S} i$.

This is the lattice path incarnation of the ground-set element $(S, \mathfrak{d}) \in \mathcal{G}_{n,c}$: the ALCD $\mathfrak{d}$ corresponds to the NILP family $\mathfrak{p}$, and the subset S records the crossing positions.

4.2 Non-intersecting vs. intersecting families

The two cases carry fundamentally different combinatorial meaning:

- $S = \emptyset$ (non-intersecting): The pair $(\emptyset, \mathfrak{p})$ is a genuine NILP family of n non-intersecting paths. Under the CPP–NILP bijection (Theorem 2), this corresponds to a *cylindric plane partition*. Sign: $+1$.
- $S \neq \emptyset$ (intersecting): The pair $(S, \mathfrak{p})$ represents a path configuration where $|S|$ crossings have occurred. The NILP family $\mathfrak{p}$ has only $n - |S|$ non-intersecting paths; the elements of S mark the levels at which paths have crossed. This does *not* correspond to a CPP. Sign: $(-1)^{|S|}$.

Every CPP of level n gives a non-intersecting family $(\emptyset, \mathfrak{p})$. But a general Q-lattice path family $(S, \mathfrak{p})$ with $S \neq \emptyset$ represents an *intersecting* configuration — and these are precisely what the involution must cancel.

4.3 What are intersecting paths?

In the CPP world, every cylindric plane partition corresponds to a non-intersecting lattice path configuration. There are no "intersecting CPPs." So where do the intersecting Q-lattice path families live?

They live in a *larger space* than the CPP space. In the multipartition representation, a CPP with profile $c = (c_0, c_1, c_2)$ and max entry n is a triple of partitions (ν^1, ν^2, ν^3) satisfying the cyclic interlacing conditions at every level $s = 1, \ldots, n$:

$$\nu^1_s \geq \nu^2_{s+c_1}, \qquad \nu^2_s \geq \nu^3_{s+c_2}, \qquad \nu^3_s \geq \nu^1_{s+c_0}. \tag{6}$$

These conditions encode the non-intersecting property: at each level s, the three lattice paths maintain their relative order on the cylinder.

A Q-lattice path family $(S, \mathfrak{p})$ with $S \neq \emptyset$ corresponds to a path configuration where the interlacing conditions (6) are *relaxed* at the levels in S:

- At levels $s \notin S$: the interlacing holds (paths maintain order $\Rightarrow$ non-intersecting).
- At levels $s \in S$: the interlacing may fail (paths cross $\Rightarrow$ intersection at level s).

Remark 7 (The enlarged path space)**.** The ground set $\mathcal{G}_{n,c}$ is the enlarged path space: the set of all path configurations on the cylinder—both non-intersecting and intersecting—indexed by $(S, \mathfrak{p})$. For $S = \emptyset$, we recover the CPP space. For $S \neq \emptyset$, we get configurations outside the CPP space: triples of partitions where the cylindric closure condition fails at the levels in S.

This parallels the classical LGV setup:

- The LGV determinant sums over *all* path families (all permutations $\sigma \in S_r$).
- The identity permutation gives non-intersecting paths.
- Other permutations give intersecting paths.
- The involution cancels intersecting paths, leaving only non-intersecting ones.

The Q-lattice path framework does the same, but the "all path families" are indexed by subsets $S \subseteq \{1, \ldots, n\}$ (recording *where* paths cross) rather than by permutations $\sigma \in S_r$ (recording *which* paths cross).

4.4 The Q-polynomial as a signed path count

The Q-polynomial is the signed generating function over all Q-lattice path families:

$$\boxed{Q_n = \sum_{(S,\mathfrak{p}) \in \mathcal{G}_{n,c}} (-1)^{|S|}\, q^{\sigma(S)+|\mathfrak{p}|}.} \tag{7}$$

The non-intersecting families ($S = \emptyset$) contribute with sign $+1$. The intersecting families ($S \neq \emptyset$) contribute with alternating signs based on the number of crossings. Warnaar's conjecture states that the positive contributions dominate.

5 The Euler factor and crossings

The crossings in a Q-lattice path family arise from the Euler factor in the E–H factorisation.

5.1 How crossings arise

The Q-polynomial comes from:

$$Q_n = (q)_n \cdot [z^n] \big((zq;q)_\infty \, \mathrm{GK}_c(z,q)\big).$$

The factor $\mathrm{GK}_c(z,q)$ counts CPPs — equivalently, *non-intersecting* NILP families. The Euler factor $(zq;q)_\infty = \prod_{i\geq 1}(1-zq^i)$ relaxes the non-intersecting condition by introducing crossings: each factor $(1-zq^i)$ either

- leaves the family unchanged ($\times 1$), or
- introduces a *crossing* at level i ($\times(-zq^i)$).

Choosing j crossing levels $1 \leq \ell_1 < \cdots < \ell_j$ contributes $(-z)^j q^{\ell_1+\cdots+\ell_j}$. Summing over all choices:

$$\sum_{1\leq \ell_1<\cdots<\ell_j} q^{\ell_1+\cdots+\ell_j} = \frac{q^{j(j+1)/2}}{(q)_j},$$

recovering Euler's identity $(zq;q)_\infty = \sum_{j\geq 0} \frac{(-1)^j z^j q^{j(j+1)/2}}{(q)_j}$.

5.2 The q-binomial decomposition in path language

Combining the Euler factor with GK_c and extracting $[z^n]$:

$$Q_n = \sum_{j=0}^{n} (-1)^j \, q^{j(j+1)/2} \begin{bmatrix} n \\ j \end{bmatrix} h_{n-j}, \tag{8}$$

where $h_m = (q)_m \cdot [z^m]\ \mathrm{GK}_c(z,q) \in \mathbb{N}[q]$.

In path language, each term has a concrete meaning:

- n path levels, of which j have crossings and $n-j$ are non-intersecting,
- $(-1)^j$: sign from j crossings,
- $q^{j(j+1)/2}$: minimum crossing weight (crossings at the lowest possible levels $1, 2, \ldots, j$),
- $\begin{bmatrix} n \\ j \end{bmatrix}$: ways to arrange j crossings among n levels, with q-weight from their positions,
- h_{n-j}: Q-weighted count of the $n-j$ non-intersecting paths.

6 The crossing involution

The sign-reversing involution ι on $\mathcal{G}_{n,c}$ ([3]) has a concrete path interpretation: it *introduces* or *resolves* crossings in a Q-lattice path family.

6.1 Introducing and resolving crossings

Definition 8 (Crossing operations)**.** Given a Q-lattice path family $(S, \mathfrak{p})$:

- **Introduce a crossing** (peel): extract a path from $\mathfrak{p}$ at level ℓ using the inverse local rule. The path collapses into a crossing at ℓ:

$$(S, \mathfrak{p}) \xrightarrow{\text{peel}} (S \cup \{\ell\},\ \mathfrak{p}').$$

  One fewer non-intersecting path, one more crossing. Sign reverses.

- **Resolve a crossing** (glue): take a crossing at position $i \in S$ and re-insert a path at level i using the forward local rule:

$$(S, \mathfrak{p}) \xrightarrow{\text{glue}} (S \setminus \{i\},\ \tilde{\mathfrak{p}}).$$

  One more non-intersecting path, one fewer crossing. Sign reverses.

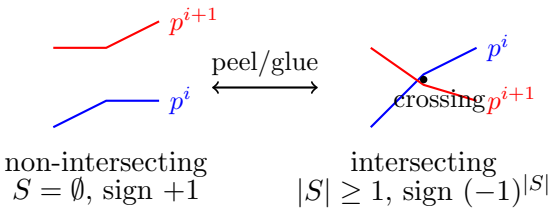
6.2 Two directions for crossings

The inverse local rule has two cascade directions:

- **Burge** (cascade up): crosses the path from below,
- **RSK** (cascade down): crosses the path from above.

The paired involution compares the Burge crossing candidate ℓ_B with the RSK crossing candidate ℓ_R and acts on whichever wins the Franklin-style comparison ([3], Section 13).

6.3 Geometry: crossing and uncrossing



The peel operation converts a non-intersecting pair of paths into an intersecting one (introducing a crossing); the glue operation resolves a crossing back into a non-intersecting configuration. The involution ι pairs these two types.

7 Positivity as cancellation of crossings

Warnaar's Conjecture 2.7 states $Q_n \in \mathbb{Z}_{\geq 0}[q]$. In the Q-lattice path framework, this is a statement about the cancellation of intersecting families.

Conjecture 9 (Path form)**.** *The signed sum* (7) *over Q-lattice path families of level n has non-negative coefficients. Equivalently: all intersecting families ($S \neq \emptyset$) can be cancelled by the sign-reversing involution ι, leaving only non-intersecting families ($S = \emptyset$).*

Proposition 10. *If every intersecting Q-lattice path family ($|S| \geq 1$) is paired by ι with a family of opposite sign, then:*

$$Q_n = \sum_{\substack{(\emptyset,\mathfrak{p}) \\ \text{fixed by } \iota}} q^{|\mathfrak{p}|} \in \mathbb{Z}_{\geq 0}[q].$$

That is: Q_n counts exactly the non-intersecting Q-weighted NILP families that survive the cancellation of all intersecting families.

Remark 11 (Why positivity is natural)**.** The path picture makes positivity intuitively plausible. An intersecting family has paths that cross, and a crossing can typically be resolved by "uncrossing" the offending pair. The non-intersecting condition constrains which crossings can occur, and for most configurations, a sign-reversing partner exists. The rare configurations that survive — the genuine non-intersecting families — contribute positively.

Remark 12 (Rank 3 constraint)**.** For rank 3, the cylinder has circumference $T = d$ and the NILP families have at most 3 paths. The crossing structure is therefore very constrained: at most two adjacent pairs of paths can cross. This small size is what makes the conjecture tractable.

8 The Q-kernel and paired crossings

For the pair set $\mathcal{P}_{a,b,k} = \mathcal{G}_{a+k} \times \mathcal{G}_{b+k}$, the NILP interpretation involves *two* Q-lattice path families coupled at cascade level k.

Definition 13 (Q-NILP pair)**.** A *Q-NILP pair* at cascade level k with strip sizes (a, b) is a pair of Q-lattice path families:

$$\big((S, \mathfrak{p}),\ (S', \mathfrak{p}')\big) \in \mathcal{G}_{a+k,c} \times \mathcal{G}_{b+k,c}.$$

The first family has level $a + k$; the second has level $b + k$. The cascade parameter k couples them.

The Q-kernel $\Phi_{a,b}(t)$ is the signed generating function of Q-NILP pairs:

$$\Phi_{a,b}(t) = \sum_{k \geq 0} t^k \sum_{\substack{(S,\mathfrak{p}) \in \mathcal{G}_{a+k} \\ (S',\mathfrak{p}') \in \mathcal{G}_{b+k}}} (-1)^{|S|+|S'|}\, q^{\sigma(S)+|\mathfrak{p}|+\sigma(S')+|\mathfrak{p}'|}. \tag{9}$$

The paired involution ι ([3], Section 13) becomes a *coupled crossing operation*: introduce a crossing in one family (Burge direction) or the other (RSK direction), comparing the two crossing candidates to decide which family to act on.

9 The LGV perspective

The Lindström–Gessel–Viennot lemma gives a determinantal expression for the generating function of NILP families.

9.1 Standard LGV

For sources $A = (a_1, \ldots, a_r)$ and sinks $B = (b_1, \ldots, b_r)$ on a directed acyclic graph, the LGV lemma states:

$$\sum_{\text{NILPs } A \to B} \prod_i \mathrm{wt}(p^i) = \det\big(h(a_i, b_j)\big)_{1 \leq i,j \leq r}, \tag{10}$$

where $h(a, b)$ is the (weighted) generating function for all paths from a to b.

For the cylinder, the LGV lemma gives:

$$\mathrm{GK}_c(z, q) = \det\big(h_c(a_i, b_j; z, q)\big), \tag{11}$$

where h_c is the cylindric path generating function.

9.2 Q-LGV

In the Q-enriched setting, each path step carries a Q-weight. The Q-enriched path generating function is:

$$h_c^Q(a, b; z, q) = \sum_{\text{paths } a \to b} \prod_{\text{steps}} Q_{a_k}(q) \cdot z^{\#\text{up-steps}} \cdot q^{\text{weight}}. \tag{12}$$

Proposition 14 (Q-LGV)**.** *The Q-weighted NILP generating function is:*

$$\sum_{Q\text{-}NILPs} \mathrm{wt}_Q \cdot z^{|\mathfrak{c}|} = \det\big(h_c^Q(a_i, b_j; z, q)\big). \tag{13}$$

The Q-enrichment replaces the scalar entries of the LGV matrix with Q-weighted path counts.

Remark 15 (LGV and crossings)**.** The LGV determinant naturally includes all path families — both non-intersecting and intersecting — with sign from the permutation. Off-diagonal permutations force paths to cross. The Q-lattice path family $(S, \mathfrak{p})$ refines this: S tracks the crossings explicitly, and the involution operates on individual crossings rather than on permutations.

9.3 LGV crossings vs. Q-NILP crossings

The crossing structure in the Q-NILP framework differs fundamentally from that in the classical LGV lemma:

	Classical LGV	**Q-NILP**
Crossing indexed by	$\sigma \in S_r$	$S \subseteq \{1, \ldots, n\}$
Meaning	which paths swap destinations	at which levels paths cross
Number of types	$r!$ (topological)	2^n (geometric)
Sign	$\mathrm{sgn}(\sigma)$	$(-1)^{\lvert S\rvert}$
Involution	swap tails at first intersection	peel/glue at first crossing level
Fixed points	NILP (identity permutation)	non-intersecting Q-families

In the LGV framework, the permutation σ determines the *topology* of the crossing: which source connects to which sink. For rank 3, there are $3! = 6$ topological types.

In the Q-NILP framework, the subset S determines the *geometry* of the crossing: at which levels along the cylinder the paths intersect. For a family of level n, there are 2^n geometric patterns.

Both decompositions express the same signed generating function. However, the Q-NILP decomposition is *finer*: it tracks individual crossing events at each level rather than the global crossing topology. This finer granularity is what makes a Franklin-style level-by-level involution possible.

Remark 16 (Why subsets, not permutations?)**.** In the classical LGV lemma on a directed acyclic graph (DAG), paths are uniquely determined by their endpoints, so the crossing structure is encoded by the endpoint permutation $\sigma \in S_r$.

On the *cylinder*, paths can wind around, and the same pair of endpoints can give rise to many different path families. The level-by-level structure of the growth diagram resolves this multiplicity: each level is a separate commutation step, and the subset S records which steps involve crossings. This is why the Q-crossing data is a subset of $\{1, \ldots, n\}$ rather than a permutation.

10 Specialisations

10.1 $d = 1$: standard NILPs

When $d = 1$, $Q_n = 1$ for all n. Every cube carries weight 1, and the Q-NILP model reduces to the standard NILP model. The LGV determinant gives the Borodin product formula, and all intersecting families cancel perfectly.

10.2 $d = 2$ (Rogers–Ramanujan): quadratic paths

For $d = 2$ with $c = (2, 0, 0)$: $Q_n = q^{n^2}$. Each cube at strip size n carries weight q^{n^2} — a quadratic function of the path position. The NILP family resembles paths in a quadratic potential.

The signed path count becomes:

$$Q_n = q^{n^2} = \sum_{(S,\mathfrak{p})} (-1)^{|S|}\, q^{\sigma(S)+|\mathfrak{p}|},$$

which states that the intersecting families cancel so completely that only a single monomial survives. This extreme cancellation reflects the Rogers–Ramanujan structure.

10.3 $d = 4$, $c = (2, 2, 0)$

Here $Q_1 = q^2 + q^3 + q^4 + q^5$: a single cube can carry four different weights. The NILP picture shows paths in a "multi-well potential" where each cube position contributes a polynomial weight. The non-negativity of Q_n for all n means that all intersecting families can be cancelled, despite the complex weight structure.

11 Connection to the pair framework

The lattice path picture gives a geometric underpinning to the pair-based involution programme of [3]:

Pair framework	**NILP picture**
Ground-set element $(S, \mathfrak{d})$	Q-lattice path family $(S, \mathfrak{p})$
ALCD $\mathfrak{d}$	Non-intersecting path family $\mathfrak{p}$
Subset S	Crossing positions
$S = \emptyset$	Non-intersecting family (= CPP)
$S \neq \emptyset$	Intersecting family (to be cancelled)
Peel (inverse rule)	Introduce a crossing
Glue (forward rule)	Resolve a crossing
Q-Burge (cascade up)	Cross from below
Q-RSK (cascade down)	Cross from above
Cascade parameter k	Number of cubes between two paths
Q-kernel $\Phi_{a,b}(t)$	Two-path interaction GF
Fixed points of ι	Non-intersecting families that survive

12 Open questions

1. **Explicit Q-LGV matrix.** Compute the Q-enriched LGV matrix for rank 3 profiles. The matrix entries $h_c^Q(a_i, b_j)$ should be expressible in terms of the Q_n and q-binomial coefficients.

2. **Crossing involution.** Make the crossing involution explicit: given an intersecting Q-lattice path family $(S, \mathfrak{p})$ with $S \neq \emptyset$, specify which crossing to resolve (or which new crossing to introduce), using the non-intersecting condition as a guide. The Burge direction (cross from below) and RSK direction (cross from above) give two options.

3. **Fixed-point characterisation.** Characterise the fixed points of the crossing involution. Are they always non-intersecting ($S = \emptyset$)? If so, Q_n counts genuine NILP families with Q-weighted cubes.

4. **Rhombus tilings.** The NILP model is dual to rhombus tilings on the cylinder. What does the Q-enrichment mean for tilings? Each rhombus type would carry a Q-weight, and the tiling count becomes a Q-polynomial.

5. **Cylinder circumference 3.** For rank 3, the cylinder has circumference $T = d$ and the NILP families have at most 3 paths with the LGV matrix being 3×3. Work out the crossing structure explicitly for small n to see the cancellation in action.

References

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